

Employment

Graduate Research Assistant	NC State University	Apr 2025 – Present
- Designed and deployed a cloud-native ML data platform on GCP (Cloud Run) integrating JBrowse 2 genome browser, enabling feature extraction and model training on 500GB+ genomic datasets for 100+ researchers. - Improved ML model accuracy by 12% through systematic hyperparameter tuning and time-series feature engineering on biological sensor data; built scikit-learn evaluation pipelines with cross-validated metrics.		
Product Engineer Intern	Isomer AI	May 2025 – May 2026
- Deployed 10+ production AI features into enterprise workflows: multi-model LLM pipelines (Claude, GPT-4o, Gemini), Temporal.io agentic workflows with MCP tooling, and Auth0 RBAC; 923 commits, 7 deployments. - Owned the full backend stack (PostgreSQL, Redis, Docker, Terraform, Sentry APM); managed CI/CD pipelines, resolved production incidents on-call, and upheld SLA across enterprise client environments.		
Software Developer Intern	Mphasis (HP Company)	Jun 2024 – Aug 2024
- Built 5+ AI-driven workflows including a health claim automation tool using OCR, NLP, and ML models, improving document processing accuracy by 25% across high-volume enterprise insurance pipelines. - Automated AWS and GCP deployments via CI/CD pipelines, integrated Grafana dashboards for real-time monitoring, and led model red-teaming to identify failure modes and improve production reliability.		
AI/ML Intern	Capgemini	Jun 2023 – Aug 2023
- Led full SDLC for a data-driven support ticket analysis system using NLP classification models, reducing average resolution time by 15% and improving first-contact resolution rates across enterprise clients. - Built distributed data pipelines using Kafka streaming for real-time ingestion and Spark + Hadoop for batch processing, cutting ticket-processing latency by 20% and improving overall data throughput.		

Education

Raleigh, NC	North Carolina State University	Aug 2024 – May 2026
- Master of Science in Computer Science - Courses: Artificial Intelligence, Operating Systems, Algorithms, Data Science, Object Oriented Development, Software Engineering, Programming Languages, Computer Networks, Distributed Systems.		
Mumbai, India	Mumbai University	Aug 2020 – May 2024
Bachelor of Technology in Computer Science; CGPA: 9.32/10. Courses: Data Warehousing, Big Data Infrastructure, DS & Algorithms, Machine Learning, Blockchain, Computer Networks, Database Systems.		

Projects

CortexQ	Kubernetes LLM Inference Platform	2026
- Built Kubernetes-native LLM inference platform: FastAPI router with EWMA latency-aware load balancing across Claude, GPT-4o, and Gemini; KEDA autoscaler scales pods 0 to N on Redis queue depth; 88 pytest tests. - Implemented a Python Kopf CRD operator reconciling Deployments, Services, and KEDA ScaledObjects; wired OpenTelemetry + Prometheus + Grafana observability stack and shipped GitOps delivery via Helm + ArgoCD.		
VectorLift	Semantic Search & Ranking Engine	2026
- Built a semantic search system on MS MARCO: BM25 (Elasticsearch), bi-encoder retrieval fine-tuned with in-batch negatives, and cross-encoder re-ranker; evaluated 6 pipeline configs with NDCG@10 and MRR@10. - Designed a Kafka-driven indexing pipeline for event-based corpus ingestion into Elasticsearch and Qdrant; served search via FastAPI with per-stage latency tracking and Streamlit experiment dashboard.		

Languages and Technologies

- Languages:** Python, TypeScript, JavaScript, Java, SQL, Bash
- AI/ML:** LLMs (Claude, GPT-4o, Gemini), MCP, Temporal.io, Kafka, scikit-learn, PyTorch, Hugging Face, RAG
- Data & Storage:** PostgreSQL, Redis, Elasticsearch, Qdrant, S3, MySQL, MongoDB, DuckDB
- Cloud & Infra:** AWS, GCP, Docker, Kubernetes, KEDA, Terraform, FastAPI, Node.js, GitHub Actions, Helm

Additional Experience and Awards

- Publications:** Co-authored CV-Based Cybersecurity Threat Detection (Springer icSoftComp 2023) and Mapify (IEEE 2022); presented to 500+ global attendees at icSoftComp international conference.
- Hackathon Winner:** Won or ranked Top 10 in 6+ national hackathons; shipped 3 full-stack MVPs end-to-end within 48-hour competitive sprints.
- GDSC Lead:** Led Google Developer Student Club; organized 4+ tech events (hackathons, conferences) reaching 500+ students.
- Technical Writing:** Published articles on LLM model collapse, AI agent security, and audio AI at kulkarniaditya.com/blog.